(Solutions) Exam III, Alternate

Calculus I, Fall 2017

Problem 1

Use Logarithmic differentiation to find $\frac{dy}{dx}$, where

$$y = \frac{e^x(2x+1)^6}{3x+2}.$$

First, we note that

$$\begin{aligned}
y &= \frac{e^x(2x+1)^6}{3x+2} \\
\ln(y) &= \ln\left(\frac{e^x(2x+1)^6}{3x+2}\right) \\
\ln(y) &= \ln\left(e^x(2x+1)^6\right) - \ln(3x+2) \\
\ln(y) &= \ln(e^x) + \ln\left((2x+1)^6\right) - \ln(3x+2) \\
\ln(y) &= x + 6\ln(2x+1) - \ln(3x+2)
\end{aligned}$$

So, it follows then that

$$\frac{d}{dx}(\ln(y)) = \frac{\frac{dy}{dx}}{y} = \frac{d}{dx}(x + 6\ln(2x+1) - \ln(3x+2))$$

That is,

$$\frac{dy}{dx} = y\frac{d}{dx}(x + 6\ln(2x+1) - \ln(3x+2))$$

Calculultating, we find

$$\begin{aligned}
\frac{dy}{dx} &= y\left(1 + 6\frac{2}{2x+1} - \frac{3}{3x+2}\right) \\
&= y\left(1 + \frac{12}{2x+1} - \frac{3}{3x+2}\right) \\
&= \frac{e^x(2x+1)^6}{3x+2}\left(1 + \frac{12}{2x+1} - \frac{3}{3x+2}\right) \\
&= \frac{e^x(2x+1)^6}{3x+2} + \frac{12e^x(2x+1)^5}{(2x+1)(3x+2)} - \frac{3e^x(2x+1)^6}{(3x+2)^2}
\end{aligned}$$

Or, you could just use the chain rule to, ultimately, get the exact same thing (but probably written in a different form, depending on how you did your simplifications).

Problem 2

Determine whether each of the following statements is true or false, giving a brief explanation in either case. (Note that all three parts relate to the Mean Value Theorem.)

Part (a)

If f is a polynomial, then between any two roots of f there is a root of f'.

Let f be a polynomial. Then, f is continuous everywhere and differentiable everywhere. Let a and b be two roots of f (that is, let a and b be such that $f(a) = f(b) = 0$). Then, Rolle's Theorem (or the Mean Value Theorem, if you prefer) applies to f on the interval $[a,b]$, and there correspondingly exists a point c in (a,b) (that is, a point between the roots a and b) such that $f'(c) = 0$. Hence, we have shown that between any two roots of a polynomial function f, there exists a root of f'. Correspondingly, the statement is true.

Part (b)

The continuous function $f(x) = 1 - |x|$ satisfies the conditions of the Mean Value Theorem (or Rolles Theorem) on the interval $[-1,1]$.

We know

$$f(-1) = 1 - |-1| = 0 = 1 - |1| = f(1)$$

So, $f(x)$ satisfies $f(-1) = f(1)$. We also know $f(x)$ is continuous (as given in the statement of the problem) on $[-1,1]$. Similarly, $f(x)$ is differentiable for all values of x except $x = 0$, so $f(x)$ is not differentiable on (-1.1). Hence, $f(x)$ does not satisfy the conditions of the Mean Value Theorem (or, equivalently, Rolle's Theorem, since $f(a) = f(b)$). Correspondingly, the statement is false.

Part (c)

If $P(t) = \frac{200t}{1+t}$ is the population of cells at time t, then there is a time t_0 where the instantaneous rate of change of the cell population equals the average rate of change over the time interval $[0,8]$.

Let us notice that $P(t)$ is a rational function, and is therefore continuous on its domain. The domain of $P(t)$ is $(-\infty,-1)\bigcup(-1,\infty)$, as the denominator of $P(t)$ is nonzero at every point other than $t = -1$. So, $P(t)$ is certianly continuous on $[0,8]$. Moreover, $P'(t) = \frac{200(1+t)-200t}{(1+t)^2}$, so $P'(t)$ is defined for all values of t other than $t = -1$. Therefore, $P(t)$ is differentiable on $[0,8]$. Thus, the Mean Value Theorem applies, and there exists a point c in $(0,8)$ such that

$$P'(c) = \frac{P(8) - P(0)}{8 - 0}.$$

That is, there exists a point c in $(0,8)$ such that $P'(c)$ is equal to the average rate of change of $P(t)$ over the time interval $[0,8]$. Correspondingly, the statement is true.

Problem 3

If the curve $y = f^{-1}(x)$ has tangent line of slope $\frac{1}{4}$ at the point $(7,5)$, find $f'(5)$.

Note that the Inverse Function Theorem (that is, Theorem 12 in your notes from Chapter 3, section 10), yields that

$$(f^{-1})'\big(f(x_0)\big) = \frac{1}{f'(x_0)}$$

for functions f which have an inverse and are differentiable on an interval I, where $f'(x_0) \neq 0$ and x_0 is a point in I.

So, as $(7,5)$ is a point on the graph of $y = f^{-1}(x)$, the point $(5,7)$ is correspondingly a point on the graph of $y = f(x)$. That is, $f(5) = 7$. So, if $y = f^{-1}(x)$ has tangent line of slope $\frac{1}{4}$ at the point $(7,5)$, then we know that

$$(f^{-1})'(7) = \frac{1}{4}.$$

But, using the Inverse Function Theorem as stated and our observation above, this means

$$(f^{-1})'\big(f(5)\big) = \frac{1}{f'(5)} = \frac{1}{4}.$$

So, solving, $f'(5) = 4$.

Problem 4

Determine the derivatives of the following functions using the rules you have learned. Do not simplify your answers.

Part (a)

$f(x) = \sin^{-1}(x^2 + x + 1)$

$$f'(x) = \frac{2x+1}{\sqrt{1-(x^2+x+1)^2}}$$

Part (b)

$g(x) = \ln\big(\ln(x)\big)$

$$g'(x) = \frac{1}{x\ln(x)}$$

Part (c)

$h(x) = \sin(3^x)$

$$h'(x) = 3^x \ln(3)\cos(3^x)$$

Problem 5

The graph of $f(x) = \sin|x|$, $2\pi < x < 2\pi$ is shown (see practice exam).

Part (a)

Where are the inflection points of f? (List their x-coordinates.)

$$\text{The inflection points of } f \text{ occur at } x = -\pi, \quad x = \pi$$

Part (b)

Where are the local maxima of f? (List their x-coordinates.)

$$\text{Local maxima of } f \text{ occur at } x = -\frac{\pi}{2}, \quad x = \frac{\pi}{2}$$

Part (c)

Where are the local minima of f? (List their x-coordinates.)

$$\text{Local minima of } f \text{ occur at } x = -\frac{3\pi}{2}, \quad x = \frac{3\pi}{2}$$

Part (d)

On which open intervals is the second derivative of f positive?

$$f''(x) > 0 \text{ on } (-2\pi, -\pi) \text{ and } (\pi, 2\pi)$$

Part (e)

On which open intervals is the second derivative of f negative?

$$f''(x) < 0 \text{ on } (-\pi, 0) \text{ and } (0, \pi)$$

Problem 6

Let $f(x) = \frac{1}{3+x^2}$. You may use the fact that

$$f'(x) = -\frac{2x}{(3+x^2)^2}; \quad f''(x) = \frac{6(x^2-1)}{(3+x^2)^3}.$$

Part (a)

On which interval(s) is f increasing?

$$f \text{ is increasing on } (-\infty, 0)$$

Part (b)

On which interval(s) is f decreasing?

$$f \text{ is decreasing on } (0, \infty)$$

Part (c)

What is the absolute maximum value of f? Where does f attain this maximum value?

$$f \text{ has an absolute maximum of } \frac{1}{3} \text{ which occurs at } x = 0.$$

Part (d)

On which interval(s) is the graph of f concave up?

The graph of f is concave up on the intervals $(-\infty, -1)$ and $(1, \infty)$

Part (e)

On which interval(s) is the graph of f concave down?

The graph of f is concave down on the interval $(-1, 1)$

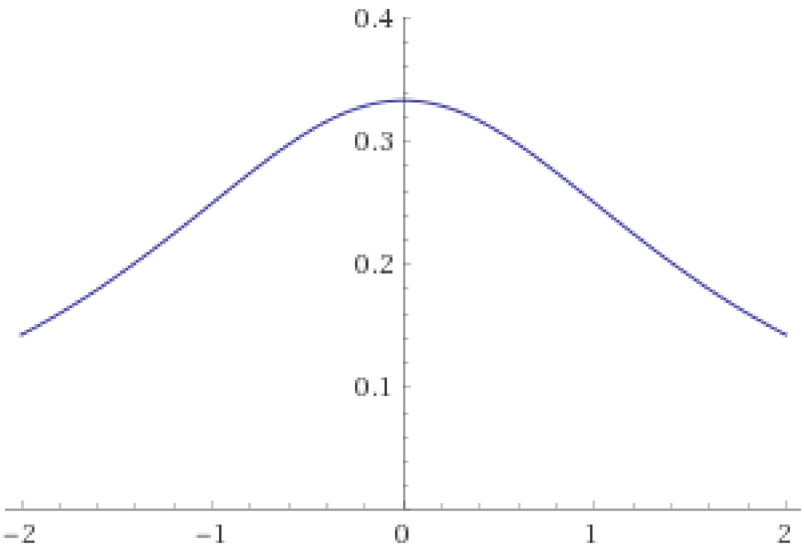
Part (f)

At which x-value(s) does the graph of f have inflection points?

The graph of f has inflection points which occur at $x = -1$ and $x = 1$

Part (g)

Using the information above, sketch the graph of f on $[-2, 2]$ using the axes provided.



Problem 7

Let $f(x) = \frac{1}{\sqrt{x}}$.

Part (a)

What is the linear approximation $L(x)$ to $f(x)$ at $x = 17$? Show your work.

$$\begin{aligned} L(x) &= \frac{-1}{2(17)^{2/3}}(x-17) + f(17) \\ &= \frac{-1}{34\sqrt{17}}x + \frac{17}{34\sqrt{17}} + \frac{1}{\sqrt{17}} \\ &= \frac{-1}{34\sqrt{17}}x + \frac{3}{2\sqrt{17}} \end{aligned}$$

Part (b)

Use your linear approximation from (a) to approximate the value of $\frac{1}{\sqrt{17}}$. Show your work and give the value of your approximation to 7 decimal places after the decimal point.

$$\begin{aligned} L(17) &= \frac{-1}{34\sqrt{17}}17 + \frac{3}{2\sqrt{17}} \\ &= \frac{1}{\sqrt{17}} \\ &\approx 0.242536 \end{aligned}$$

Notice, the linear approximation $L(x)$ yields an *exact* approximation of $\frac{1}{\sqrt{17}}$ at $x = 17$. We can clearly see this, as

$$L(17) = \frac{-1}{2(17)^{2/3}}(17-17) + f(17) = 0 + f(17) = \frac{1}{\sqrt{17}}.$$

So, the 7-decimal-digit approximation is just the first 7 decimal digits of $\frac{1}{\sqrt{17}}$. That is, we didn't really do anything. I'm not sure why the person who wrote this chose to make this the question, honestly.

Problem 8

A small balloon is released at a point 160 feet away from an observer who is on level ground. If the balloon goes straight up at a speed of 8 feet per second, how fast is the distance from the observer to the balloon increasing when the balloon is 40 feet high?

We know $L^2 = y^2 + 160^2$. So,

$$\begin{aligned}
\frac{d}{dt}\left(L^2\right) &= \frac{d}{dt}\left(y^2 + 160^2\right) \\
2L\frac{dL}{dt} &= 2y\frac{dy}{dt} \\
\frac{dL}{dt} &= \frac{2y\frac{dy}{dt}}{2L}
\end{aligned}$$

As $y = 40$, $L = \sqrt{40^2 + 160^2} = 40\sqrt{17}$ and $\frac{dy}{dt} = 8$, we may see that

$$\frac{dL}{dt} = \frac{2(40)(8)}{2(40\sqrt{17})} = \frac{8}{\sqrt{17}}.$$

Problem 9

Of all boxes with square base and a volume of 100 m^3, which one has the minimum surface area? Find the minimum surface area and provide the dimensions of the box attaining this minimum. (Justify why your dimensions give the absolute minimum surface area.) Note that this is a closed box with six sides.

First, we find our constraint and our objective function.

$$\text{Constraint: } x^2y = 100 \ \ m^3, \quad \text{Objective Function: } \mathrm{S}(x,y) = 4(xy) + 2x^2$$

Solving the constraint, we find

$$y = \frac{100}{x^2}.$$

This produces the a objective function subject to the constraint, given by

$$\mathrm{S}(x) = \mathrm{S}\left(x, \frac{100}{x^2}\right) = 4x\frac{100}{x^2} + 2x^2 = \frac{400}{x} + 2x^2$$

Now, we optimize. First, we find the critical points. While the derivative of $\mathrm{S}(x)$ will be undefined at $x = 0$, our function $\mathrm{S}(x)$ is also undefined at $x = 0$, so we may disregard this as a critical point. Thus, we now need only find the zeros of the derivative of $\mathrm{S}(x)$:

$$\begin{aligned} \mathrm{S}'(x) &= 0 \\ 4x - \frac{400}{x^2} &= 0 \\ 4x &= \frac{400}{x^2} \\ 4x^3 &= 400 \\ x &= \sqrt[3]{\frac{400}{4}} = \sqrt[3]{100}. \end{aligned}$$

So, of all boxes with square base and a volume of 100 m^3, the minimum surface area is achieved when

$$x = \sqrt[3]{100} \text{ and } y = \frac{100}{\left(\sqrt[3]{100}\right)^2} = \sqrt[3]{100}.$$

Indeed, this surface area is just

$$\mathrm{S}(\sqrt[3]{100}) = \frac{400}{\sqrt[3]{100}} + 2\left(\sqrt[3]{100}\right)^2 = 60\sqrt[3]{10}.$$